Civil Engineering

Calculus and Differential Methods

Assignment 1

Instructions: Answer all questions. Show formula, substitution, calculation, units and technical interpretation.

Section A: Formula and Concept Questions

1. Explain the use of the formula below and identify the meaning of the symbols. (4 marks)

$$\frac{d}{dx}(ax^n)=anx^{n-1}$$

2. Explain the use of the formula below and identify the meaning of the symbols. (4 marks)

$$\int ax^n\,dx=\frac{ax^{n+1}}{n+1}+C$$

3. Explain the use of the formula below and identify the meaning of the symbols. (4 marks)

$$\frac{dy}{dx}+P(x)y=Q(x)$$

4. Explain the use of the formula below and identify the meaning of the symbols. (4 marks)

$$x_{n+1}=x_n-\frac{f(x_n)}{f'(x_n)}$$

5. Explain the use of the formula below and identify the meaning of the symbols. (4 marks)

$$\int_a^b y\,dx\approx\frac{h}{2}\left[y_0+y_n+2\sum y_i\right]$$

Section B: Applied Problems

6. Solve a civil engineering calculation problem involving road profiles. Show all working and write one technical interpretation. (10 marks)

7. Solve a civil engineering calculation problem involving flow accumulation. Show all working and write one technical interpretation. (10 marks)

8. Solve a civil engineering calculation problem involving settlement rate. Show all working and write one technical interpretation. (10 marks)

9. Solve a civil engineering calculation problem involving area estimation. Show all working and write one technical interpretation. (10 marks)

10. Solve a civil engineering calculation problem involving numerical solving. Show all working and write one technical interpretation. (10 marks)

11. Solve a civil engineering calculation problem involving road profiles. Show all working and write one technical interpretation. (10 marks)

12. Solve a civil engineering calculation problem involving flow accumulation. Show all working and write one technical interpretation. (10 marks)

13. Solve a civil engineering calculation problem involving settlement rate. Show all working and write one technical interpretation. (10 marks)

14. Solve a civil engineering calculation problem involving area estimation. Show all working and write one technical interpretation. (10 marks)

15. Solve a civil engineering calculation problem involving numerical solving. Show all working and write one technical interpretation. (10 marks)

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17. Solve a civil engineering calculation problem involving flow accumulation. Show all working and write one technical interpretation. (10 marks)

18. Solve a civil engineering calculation problem involving settlement rate. Show all working and write one technical interpretation. (10 marks)

19. Solve a civil engineering calculation problem involving area estimation. Show all working and write one technical interpretation. (10 marks)

20. Solve a civil engineering calculation problem involving numerical solving. Show all working and write one technical interpretation. (10 marks)

References and Learning Sources

These learner materials are original notes prepared for Mr Mwirigi Mathematics Training Hub. The topic sequencing and level are informed by standard engineering mathematics references and the uploaded curriculum/OS materials. No textbook pages or solution-manual pages have been reproduced.

1. John Bird, Engineering Mathematics, Newnes / Elsevier.

2. K. A. Stroud and Dexter J. Booth, Engineering Mathematics, Palgrave Macmillan.

3. Uploaded KNEC solved mathematics materials for examination-style orientation.

4. Civil Engineering, Building Technology and Land Survey curriculum and occupational-standard materials used for local alignment.